

PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

Anonymous ACL submission

Abstract

Enterprises increasingly use property graphs and Cypher to analyze fraud, risk, identity, customer, and operational data. Public Text2Cypher benchmarks rarely reflect private schemas, terminology, access patterns, or difficulty distributions. We present PIPE-Cypher, a local-model pipeline for generating organization-specific natural-language-to-Cypher benchmarks. PIPE-Cypher combines schema introspection, reverse-query grounding, constrained Cypher generation, deterministic read-only and schema validation, execution feedback, repair, diversity controls, and LLM-judge review. The system is designed for industry settings where benchmark generation must avoid paid APIs, preserve data governance, and remain repeatable as graphs evolve.

1 Introduction

Property graphs encode enterprise relationships directly: account transfers, identity entitlements, access paths, customer interactions, and fraud rings. Cypher is a common query language for these graphs. As LLMs become natural-language interfaces for graph analytics, organizations need reliable benchmarks for natural-language-to-Cypher systems on their own schemas.

Static public benchmarks are insufficient for this setting. They cannot contain private labels, relationship types, categorical values, or domain-specific wording, and they do not track schema evolution. PIPE-Cypher addresses this gap by generating private, repeatable, execution-grounded NL-Cypher benchmarks.

2 Related Work

Recent Text2Cypher resources, including Mind the Query (Chauhan et al., 2025), SyntheT2C (Zhong et al., 2025), Auto-Cypher (Tiwari et al., 2025), and Text2Cypher (Ozsoy et al., 2025), show that

high-quality Cypher data requires more than asking an LLM for query pairs. These works motivate schema grounding, execution checks, synthetic generation, verification, and complexity-aware evaluation. PIPE-Cypher differs by targeting private enterprise benchmark generation as the primary artifact: organizations bring their own graph, run local models, enforce Cypher constraints, and receive an executable benchmark with diversity and judge metadata.

Text-to-SQL benchmarks such as Spider 2.0 (Lei et al., 2024) and BIRD (Li et al., 2023) motivate realistic database tasks, execution-based evaluation, and enterprise workflow complexity. CIKM AutoQuery (Zheng et al., 2024) motivates strategy-level workload generation analysis and separating workload quality from model quality. LDBC FinBench (Qi et al., 2023) and SNB (Püroja et al., 2023) provide graph workloads suitable for industry-oriented evaluation.

For generated benchmark quality, PIPE-Cypher combines text-generation diversity diagnostics with text-to-query structure metrics. Distinct-n (Li et al., 2016) and self-BLEU-style redundancy checks (Zhu et al., 2018) capture lexical repetition, while schema coverage, query-signature diversity, structural feature rates, and normalized entropy over graph/category/difficulty cells capture properties that matter for executable Cypher benchmarks.

3 Method

PIPE-Cypher has five stages: schema profiling, template generation, reverse Cypher grounding, constrained Cypher generation and repair, deterministic validation and execution, and LLM-judge review. Deterministic checks enforce read-only safety, syntax shape, schema-visible labels and properties, explicit relationship types, observed relationship directions, schema-provided categorical values, and non-empty execution where required. A

lightweight Cypher analyzer extracts return aliases, variables, labels, relationship observations, risky constructs, and rewrite skip reasons so normalization is auditable rather than a silent string edit. The direction gate interprets both outgoing and incoming Cypher arrow syntax before schema checking, and rejects undirected relationship patterns so directionality is an executable constraint rather than only a prompt instruction.

4 Implementation

The implementation is a Python package with Neo4j as the experimental backend. The paper frames the contribution around Cypher and property graphs rather than a single vendor. Local models are served through a vLLM/OpenAI-compatible endpoint on ds-serv6, using Qwen3.5 models for generation and judging.

The intended full-quality model is Qwen3.5-35B-A3B, but our June 1, 2026 live evidence uses the cached Qwen3.5-9B fallback. The 35B-A3B weights were staged on ds-serv6 under root-backed storage. A serving-capacity check estimated that the staged weights require four A5000 GPUs under our vLLM budget, while only one GPU was safely free in the live snapshot, so the reported generation and downstream results use the 9B fallback.

The built-in FinBench profile is grounded in the public datagen snapshot export and includes typed node and relationship properties. Live schema inspection also records low-cardinality string values as categorical constraints for prompting and validation. The generated Neo4j import script uses uniqueness constraints for nodes and creates, rather than merges, transaction relationships so repeated account-to-account events are preserved. The SNB reference profile is grounded in the official Neo4j/Cypher headers and read-query files.

For generation, PIPE-Cypher uses seeded workload templates with deterministic reverse-binding queries and a mixed mode that prepends proven workload seeds to LLM-proposed templates. Every run records accepted and rejected candidates for yield analysis. Retrieved few-shot examples replace graph-specific values with typed placeholders, preserving query structure while reducing tenant-value leakage and memorized entity reuse. A dependency-light value grounder adds typed prompt annotations for categorical values and reverse-bound entities, covering punctuation variants, possessives, plurals, synonyms, name par-

tials, and small typos.

For LLM-judge review, the implementation slices schema context to the labels, relationship types, and properties used by the candidate query. This preserves validation against the full schema while keeping local 9B fallback-model prompts within context limits on larger schemas.

The deterministic Cypher layer borrows from production lessons in the BalkanID Cypher system: schema-only prompting, exact matching, relationship direction discipline, `RETURN DISTINCT`, reserved variable rejection, categorical values, required contextual return columns, fuzzy value annotations, placeholderized retrieval examples, and parser-aware rewrite boundaries. PIPE-Cypher records parser-style structure features and skips rewrites for risky constructs such as `UNION`, `CALL`, `UNWIND`, `WHERE EXISTS`, multiple `WHERE` clauses, or reserved variables.

We also estimate seed-template capacity before full generation. This check caught and fixed a scale blocker: reverse-binding execution was hard-capped to 10 rows, and no-slot negation/ranking seeds could not support full category targets. PIPE-Cypher now uses the configured binding limit and includes slotted negation and ranking seeds for both graphs.

5 Experiments

The generated benchmark contains 3,000 accepted examples: 2,000 from LDBC FinBench and 1,000 from LDBC SNB. Categories include simple retrieval, complex retrieval, simple aggregation, complex aggregation, boolean existence, negation/difference, path/temporal transaction, and ranking/top-k.

6 Results

The full live run produced 3,000 accepted examples from 4,777 candidates using local Qwen3.5-9B for generation and judging. Category-specific recovery top-ups filled the only under-target categories from the initial sequential run. Every exported example passed read-only, syntax, schema, execution, non-empty result, and judge gates.

The accepted full-run records were exported into a benchmark package with stable example identifiers, train/dev/test splits, result samples, gate metadata, aggregate statistics, and a manifest hash for artifact tracking. The export contains 3,000 accepted examples: 2,000 FinBench, 1,000 SNB, and

Gate	Evidence checked	Failure mode caught
Read-only safety	blocked Cypher write/admin tokens	destructive or operational queries
Schema validity	labels, relationship types, properties, categorical values	hallucinated graph vocabulary or values
Direction validity	observed relationship directions	reversed or invalid traversals
Cypher analysis and normalization	structure extraction, rewrite safety, RETURN DISTINCT	unsafe rewrites or duplicate/noisy rows
Question constraints	quoted values use exact literals	fuzzy matching of exact user values
Execution	query runs and returns rows	syntactic validity without answerability
LLM judge	ambiguity, semantic alignment, schema use	executable but semantically weak examples

Table 1: PIPE-Cypher candidate acceptance gates.

Graph	Target/category	Binding limit	Min seed capacity	All categories meet target
FinBench	250	300	300	Yes
SNB	125	200	200	Yes

Table 2: Built-in seed-template capacity after removing the reverse-binding 10-row cap and adding slotted negation/ranking seeds. Capacity is a theoretical scale-readiness check; final examples still must pass validation, execution, diversity, and judge gates.

Setting	Value
Accepted examples	3,000
Primary graph	LDBC FinBench, 2,000 examples
Secondary graph	LDBC SNB, 1,000 examples
Categories	8 balanced categories, 375 each
Generation/judge model	Local Qwen3.5-9B fallback
Execution backend	Neo4j Community, two live databases
Judge audit packet	80 sampled accepted/rejected pairs

Table 3: Full live experimental setup used for the generated benchmark artifact.

375 accepted examples in every planned category.

We report diversity as a diagnostic, not only as a design target. The full export has perfect category balance and near-perfect difficulty balance, but the Qwen3.5-9B fallback run still has low query-signature diversity because seeded graph-grounded templates are doing substantial work. This is useful evidence for industry deployment: the artifact is balanced and executable, while the appendix makes template concentration visible rather than hiding it.

The full-run failure taxonomy is concentrated in diversity/duplicate controls during recovery and empty execution results; only 2/4,777 candidates were schema-invalid after deterministic Cypher governance.

For judge calibration, the released artifacts include an 80-row audit packet sampled from accepted and rejected full-run candidates, with a 40/40 judge accept/reject split and coverage over both graphs and all eight categories. The calibration tooling tracks graph, category, difficulty, strategy, and judge-decision coverage, and reports

Graph	Candidates	Accepted	Acceptance	Categories at target
FinBench	3,404	2,000	0.588	8/8
SNB	1,373	1,000	0.728	8/8
Total	4,777	3,000	0.628	16/16

Table 4: Full live generation with local Qwen3.5-9B. Candidate counts include the initial sequential run and category-specific recovery top-ups.

Export artifact	Examples	FinBench	SNB	Train	Dev	Test
Live full benchmark	3,000	2,000	1,000	2,408	296	296

Table 5: Accepted live full benchmark package with stable IDs, gate metadata, result samples, statistics, and manifest hash 8bc79a53a06b291a.

agreement, precision/recall, specificity, balanced accuracy, and Cohen’s kappa once human labels are filled. The current draft treats these labels as pending human review rather than as a generation gate.

Finally, we ran a downstream Text2Cypher evaluation using local Qwen3.5-9B on the exported full test split. The model saw schema text and the natural-language question, generated Cypher, and was evaluated by live execution against the corresponding FinBench or SNB database. The result verifies that the exported artifact can drive downstream model evaluation and gives a difficulty signal: the zero-shot 9B baseline solves simple aggregation examples but struggles on more operational categories.

7 Industry Use

PIPE-Cypher is intended to be rerun when an enterprise graph changes. The generated artifact records the schema snapshot, graph profile, model identifier, validation gates, execution samples, judge scores, difficulty features, and source run for every example. Long-running jobs can be monitored from JSONL records and recovered with category-specific top-up runs that reject questions accepted

Artifact property	Value
Categories	8 balanced categories
FinBench/category	250
SNB/category	125
Difficulty split	1,521 easy / 1,479 medium
Unique labels / rel. types	7 / 9
Read/syntax/schema/exec/judge gates	3,000/3,000

Table 6: Distribution and gate summary for the exported full benchmark artifact.

Split	Examples	Parse	Schema	Exec. success	Exec. acc.	Answer F1
Live full test	296	0.959	0.905	0.622	0.189	0.189

Table 7: Downstream Text2Cypher evaluation for local Qwen3.5-9B on the exported full benchmark test split.

in earlier runs. This design supports private benchmark refreshes without exposing schemas or values to paid APIs, while still leaving audit hooks for data owners to sample accepted and rejected examples.

8 Conclusion

PIPE-Cypher provides a practical path for enterprises to create private, executable, balanced NL-to-Cypher benchmarks. The pipeline combines schema introspection, constrained generation, deterministic validation, execution feedback, diversity control, and automated judge review.

Limitations

Execution validity does not guarantee semantic correctness. LLM judges can over-accept plausible but wrong examples, so PIPE-Cypher releases a frozen post-hoc audit packet and requires complete human labels before reporting judge-human agreement. Generated artifacts may contain private schema details or sampled values, so organizations should apply normal data governance controls.

Ethics Statement

PIPE-Cypher is designed for private benchmark generation under local-model deployment constraints. Organizations using it should restrict benchmark artifacts to authorized users, review sampled values for sensitive content, and document whether judge calibration relied on human annotators.

References

Vashu Chauhan, Shobhit Raj, Shashank Mujumdar, Avirup Saha, and Anannay Jain. 2025. [Mind the query: A benchmark dataset towards Text2Cypher](#)

[task](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1890–1905, Suzhou, China. Association for Computational Linguistics.

Fangyu Lei, Jixuan Chen, Yuxiao Ye, Ruisheng Cao, Dongchan Shin, Hongjin Su, Zhaoqing Suo, Hongcheng Gao, Wenjing Hu, Pengcheng Yin, Victor Zhong, Caiming Xiong, Ruoxi Sun, Qian Liu, Sida Wang, and Tao Yu. 2024. [Spider 2.0: Evaluating language models on real-world enterprise text-to-SQL workflows](#). *Preprint*, arXiv:2411.07763.

Jinyang Li, Binyuan Hui, Ge Qu, Jiaxi Yang, Binhua Li, Bowen Li, Bailin Wang, Bowen Qin, Rongyu Cao, Ruiying Geng, Nan Huo, Xuanhe Zhou, Chenhao Ma, Guoliang Li, Kevin C. C. Chang, Fei Huang, Reynold Cheng, and Yongbin Li. 2023. [Can LLM already serve as a database interface? a BIG bench for large-scale database grounded text-to-SQLs](#). In *Advances in Neural Information Processing Systems*.

Jiwei Li, Michel Galley, Chris Brockett, Jianfeng Gao, and Bill Dolan. 2016. [A diversity-promoting objective function for neural conversation models](#). In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 110–119, San Diego, California. Association for Computational Linguistics.

Makbule Gulcin Ozsoy, Leila Messallem, Jon Besga, and Gianandrea Minneci. 2025. [Text2Cypher: Bridging natural language and graph databases](#). In *Proceedings of the Workshop on Generative AI and Knowledge Graphs*, pages 100–108, Abu Dhabi, UAE. International Committee on Computational Linguistics.

David P. ñoroja, Jack Waudby, Peter Boncz, and G. ’abor Sz. 2023. [The LDBC social network benchmark interactive workload v2: A transactional graph query benchmark with deep delete operations](#). *Preprint*, arXiv:2307.04820.

Shipeng Qi, Heng Lin, Zhihui Guo, G. ’abor Sz. 2023. [The LDBC financial benchmark](#). *Preprint*, arXiv:2306.15975.

Aman Tiwari, Shiva Krishna Reddy Malay, Vikas Yadav, Masoud Hashemi, and Sathwik Tejaswi Madhusudhan. 2025. [Auto-cypher: Improving LLMs on cypher generation via LLM-supervised generation-verification framework](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2: Short Papers*, pages 623–640, Albuquerque, New Mexico. Association for Computational Linguistics.

Metric	Value
Question Distinct-1	0.041
Question Distinct-2	0.088
Question self-BLEU-2 (sampled)	0.826
Unique query-signature ratio	0.010
Top query-signature share	0.083
Category normalized entropy	1.000
Graph-category normalized entropy	0.980
Difficulty normalized entropy	1.000
Label coverage	0.412
Relationship-type coverage	0.375
Property-name coverage	0.407
Aggregation / negation / ordering rates	0.500 / 0.125 / 0.125

Table 8: Diversity diagnostics for the full exported benchmark. Distinct-n follows text-generation usage; self-BLEU is lower when questions are less redundant.

Failure bucket	Count	Share of rejected
Diversity/duplicate control	1,335	0.751
Empty result	374	0.210
Judge semantic reject	66	0.037
Schema invalid	2	0.001

Table 9: Failure taxonomy over full-run generation candidates before benchmark export. Accepted examples are excluded from the bucket shares.

Xiuwen Zheng, Arun Kumar, and Amarnath Gupta. 2024. [Generating cross-model analytics workloads using LLMs](#). In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, pages 4303–4307. ACM.

Zijie Zhong, Linqing Zhong, Zhaoze Sun, Qingyun Jin, Zengchang Qin, and Xiaofan Zhang. 2025. [SyntheT2C: Generating synthetic data for fine-tuning large language models on the Text2Cypher task](#). In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 672–692, Abu Dhabi, UAE. Association for Computational Linguistics.

Yaoming Zhu, Sidi Lu, Lei Zheng, Jiaxian Guo, Weinan Zhang, Jun Wang, and Yong Yu. 2018. [Taxygen: A benchmarking platform for text generation models](#). In *The 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1097–1100. ACM.

A Additional Results

B Claim–Evidence Map

This section maps the main paper claims to the strongest current evidence and to the remaining risks. This is intended as a reviewer-facing audit surface: claims with pending evidence remain marked as such rather than being folded into the main results.

1. **Claim 1.** PIPE-Cypher can generate a private, executable NL-to-Cypher benchmark

Setting	Graph	Records	Accepted	Acceptance	Categories at target
Unconstrained LLM	FinBench	0	0	0.000	0/8
Unconstrained LLM	SNB	0	0	0.000	0/8
Reverse-only	FinBench	400	400	1.000	8/8
Reverse-only	SNB	402	400	0.995	8/8
Validators+repair	FinBench	400	400	1.000	8/8
Validators+repair	SNB	402	400	0.995	8/8
No retrieval	FinBench	429	400	0.932	8/8
No retrieval	SNB	402	400	0.995	8/8
No rewrite	FinBench	423	400	0.946	8/8
No rewrite	SNB	402	400	0.995	8/8
No LLM judge	FinBench	401	400	0.998	8/8
No LLM judge	SNB	406	400	0.985	8/8
Full PIPE-Cypher	FinBench	416	400	0.962	8/8
Full PIPE-Cypher	SNB	402	400	0.995	8/8

Table 10: Live target-50 ablation evidence with local Qwen3.5-9B. Each graph run targets 50 accepted examples per category.

Setting	Graph	Read-only	Syntax	Schema	Exec.	Judge
Unconstrained LLM	FinBench	0.000	0.000	0.000	0.000	0.000
Unconstrained LLM	SNB	0.000	0.000	0.000	0.000	0.000
Reverse-only	FinBench	1.000	1.000	1.000	1.000	1.000
Reverse-only	SNB	1.000	1.000	0.995	0.995	0.995
Validators+repair	FinBench	1.000	1.000	1.000	1.000	1.000
Validators+repair	SNB	1.000	1.000	0.995	0.995	0.995
No retrieval	FinBench	1.000	1.000	0.995	0.995	0.932
No retrieval	SNB	1.000	1.000	0.995	0.995	0.995
No rewrite	FinBench	1.000	1.000	1.000	1.000	0.946
No rewrite	SNB	1.000	1.000	0.995	0.995	0.995
No LLM judge	FinBench	1.000	1.000	1.000	1.000	0.998
No LLM judge	SNB	1.000	1.000	0.995	0.995	0.985
Full PIPE-Cypher	FinBench	1.000	1.000	1.000	1.000	0.962
Full PIPE-Cypher	SNB	1.000	1.000	0.995	0.995	0.995

Table 11: Quality-gate rates for the live target-50 ablation suite. Rates are computed over all generated records in each graph/setting.

over enterprise-style property graphs using local models.

Evidence. The full Qwen3.5-9B fallback run exported 3,000 accepted examples: 2,000 FinBench and 1,000 SNB, with 375 examples in every planned workload category and all accepted examples passing read-only, syntax, schema, execution, non-empty-result, and judge gates.

Key artifacts. benchmark export; manifest snapshot; paper table: benchmark export; paper table: full results

Status. Supported by live 9B fallback evidence; 35B target run is blocked by current GPU capacity.

Risk. Generation quality may change with the target 35B model or with private enterprise schemas beyond FinBench/SNB.

2. **Claim 2.** Cypher-specific governance makes generated benchmark candidates safer and more auditable than prompt-only generation.

Evidence. The pipeline validates read-only safety, schema-visible labels/properties/relationships, observed relationship direction, categorical values, con-

Audit packet property	Value
Rows	80
Judge accept / reject	40 / 40
FinBench / SNB rows	48 / 32
Easy / medium rows	26 / 54
Labeled rows	0
Calibration status	unlabeled

Table 12: Post-hoc judge calibration packet coverage. Human labels are pending until the packet is complete and are not used as a generation gate.

Metric	Point	95% CI	SE	N
Parse valid	0.959	[0.936, 0.980]	0.012	296
Schema valid	0.905	[0.872, 0.939]	0.017	296
Execution success	0.622	[0.564, 0.676]	0.028	296
Execution accuracy	0.189	[0.145, 0.233]	0.022	296
Answer F1	0.189	[0.149, 0.236]	0.023	296

Table 13: Bootstrap uncertainty for downstream Text2Cypher evaluation on the full exported test split. Intervals use row-level resampling with a fixed seed and are intended for appendix reporting.

textual return columns, parser-style structure, execution success, and LLM-judge review. In the full run, only 2 of 4,777 candidates were schema-invalid after deterministic governance, and all 3,000 exported examples passed the deterministic and judge gates.

Key artifacts. code: validator.py; code: cypher_parser.py; code: question_constraints.py; snapshot: failure taxonomy; +1 more

Status. Supported by code, tests, and full-run failure taxonomy.

Risk. Semantic correctness still depends on execution evidence and judge review; human audit labels are pending.

- Claim 3.** Reverse grounding and structured workload seeds are designed to improve reliability under local-model constraints.

Evidence. The target-50 FinBench/SNB ablation suite completed 14/14 graph/variant cells, reached every category target for all non-empty/non-unconstrained runs, and passed the paper-readiness audit with logs, model IDs, code revision, run summaries, and gate-rate availability. Appendix-only tables plus

yield and gate-rate figures report the audited target-50 evidence while the target-100 follow-up and seeded target-50 repeat continue for stronger final reliability and variance claims.

Key artifacts. script: run live ablation suite; script: monitor remote ablation suite; script: monitor remote ablation queue; script: collect remote ablation suite; +9 more

Status. Supported by audited target-50 appendix evidence; target-100 and seeded target-50 repeat runs are still pending for stronger final reliability claims.

Risk. A single target-50 suite is the minimum publishable ablation scale, not the strongest possible evidence. If target-100 or repeated-seed runs complete, final claims should use those sensitivity results instead of relying only on target-50.

- Claim 4.** The benchmark controls category and difficulty balance while exposing residual diversity limits.

Evidence. The full export has perfect category balance, planned 2:1 FinBench/SNB graph mix, and a near-even easy/medium split. Diversity diagnostics report Distinct-n, sampled self-BLEU-2, query-signature diversity, normalized entropy, schema coverage, and structural feature rates, making template concentration visible instead of hidden.

Key artifacts. snapshot: diversity report; paper table: diversity; paper figure: diversity diagnostics; paper figure: full export distribution

Status. Supported by full-export statistics and diversity diagnostics.

Risk. Low query-signature diversity in the 9B fallback run remains a limitation and ablation target.

- Claim 5.** The generated benchmark is useful for downstream Text2Cypher evaluation.

Evidence. A zero-shot local Qwen3.5-9B downstream evaluation over the 296-example full test split reports parse validity, schema validity, execution success, execution accuracy, answer F1, per-category breakdowns, and fixed-seed bootstrap uncertainty intervals. The model reaches 0.189 execution accuracy with a 95% bootstrap interval of [0.145,

442	0.233] and 0.622 execution success with in-	qwen35b capacity snapshot 20260601; +1	490
443	interval [0.564, 0.676], showing the benchmark	more	491
444	can discriminate easy aggregation from harder	<i>Status.</i> Documented blocker; current paper	492
445	operational categories.	results are explicitly reported as Qwen3.5-9B	493
446	<i>Key artifacts.</i> downstream summary; snap-	fallback evidence.	494
447	shot: downstream uncertainty; research note:	<i>Risk.</i> A later 35B run may change yield, diver-	495
448	downstream evaluation; paper table: down-	sity, judge behavior, and downstream conclu-	496
449	stream evaluation; +3 more	sions.	497
450	<i>Status.</i> Supported by live downstream evalua-		
451	tion against FinBench/SNB databases, with	C Prompt Contracts	498
452	bootstrap uncertainty computed from row-	PIPE-Cypher treats prompts as versioned imple-	499
453	level outputs.	mentation artifacts. The list summarizes the prompt	500
454	<i>Risk.</i> Only one downstream model has been	contracts used for generation, repair, judging, and	501
455	evaluated so far.	downstream evaluation; hashes fingerprint the full	502
456		prompt constants in the codebase.	503
457	6. Claim 6. Automated LLM-judge review can	1. Template generation. Stage: Workload pro-	504
458	replace human-in-the-loop generation gates	posal. SHA-256: 10b25b.	505
459	while still exposing judge risk.	<i>Contract.</i> Schema-only labels, relationships,	506
460	<i>Evidence.</i> The pipeline uses deterministic val-	properties, and categorical values; Realistic	507
461	idation plus LLM-judge review as the gen-	enterprise analyst wording; At most two typed	508
462	eration gate. A post-hoc 80-row judge au-	slots and JSON-only output	509
463	dit packet preserves 40 judge accepts, 40		
464	judge rejects, both graphs, and all eight cate-	2. Reverse binding. Stage: Graph grounding.	510
465	gories. The analyzer reports agreement, preci-	SHA-256: 48e949.	511
466	sion/recall, specificity, balanced accuracy, and	<i>Contract.</i>	512
467	Cohen’s kappa only after the packet is com-	Read-only	513
468	pletely human-labeled; partial labels are not	MATCH/WHERE/RETURN	514
469	treated as paper-ready calibration.	DISTINCT/LIMIT only; Slot variables named	515
470	<i>Key artifacts.</i> code: judge.py; code: calibra-	exactly as requested; Forward relationship	516
471	tion.py; code: judge_audit_packet.py; snap-	directions from the schema	
472	shot: judge audit packet v2; +2 more	3. Cypher generation. Stage: Candidate query.	517
473	<i>Status.</i> Pipeline gate is implemented; audit	SHA-256: 61c557.	518
474	packet coverage is supported; judge-human	<i>Contract.</i> Only schema-visible constructs and	519
475	agreement remains unreported until all audit	observed directions; RETURN DISTINCT for	520
476	rows are labeled.	set returns and exact equality for quoted val-	521
477	<i>Risk.</i> Judge-human agreement is not yet	ues; Context columns, categorical hints, place-	522
478	known because human labels are still blank.	holderized retrieval, and no writes	523
479		4. Repair. Stage: Validation feedback. SHA-	524
480	7. Claim 7. The intended 35B local genera-	256: 56c419.	525
481	tion/judge path is staged but not currently	<i>Contract.</i> Preserve question intent while fix-	526
482	runnable under observed shared-GPU con-	ing validation or execution issues; Keep query	527
483	straints.	read-only and schema-grounded; Return only	528
484	<i>Evidence.</i> Qwen3.5-35B-A3B is staged on ds-	corrected Cypher	529
485	serv6, but the capacity checker reports 68,573		
486	MiB of weights, four required A5000 GPUs	5. LLM judge. Stage: Quality gate. SHA-256:	530
487	under the conservative vLLM budget, and	073938.	531
488	only one safe GPU in the latest snapshot.	<i>Contract.</i> Inputs include question, Cypher,	532
489	<i>Key artifacts.</i> script: check vllm capacity;	schema slice, execution rows, and validation	533
	script: render vllm capacity snapshot; re-	summary; Strict JSON scores for ambiguity,	534
	search note: model availability; research note:		

535 semantic alignment, schema use, and diffi-
 536 culty; Pass only useful, unambiguous enter-
 537 prise benchmark examples

538 6. **Downstream Text2Cypher.** Stage: Model
 539 evaluation. SHA-256: 4c07ff.

540 *Contract.* Read-only Cypher only; Schema-
 541 visible constructs and exact direction preser-
 542 vation; RETURN DISTINCT, count/ranking
 543 rules, and no explanations

544 D Representative Accepted Examples

545 The examples below are selected in stable identifier
 546 order from the tracked full-export snapshot, one per
 547 graph/category cell when available. They show the
 548 natural-language question, accepted Cypher, struc-
 549 tural tags, gate status, and a bounded execution-
 550 result sample.

551 1. **FinBench / Boolean Existence / easy.** *Ques-*
 552 *tion:* Does account '187743809466009406'
 553 have any outgoing transfer?

```
554 MATCH (src:Account {accountId:  
555 '187743809466009406'})  
556 -[:TRANSFER_TO]->  
557 (:Account)  
558 RETURN DISTINCT COUNT(DISTINCT src)  
559 > 0  
560 AS HasOutgoingTransfer
```

561 *Structure:* single_hop, aggregation.

562 *Relationships:* TRANSFER_TO.

563 *Gates:* RO/Syn/Schema/Exec/Judge.

564 *Result sample:* {HasOutgoingTransfer: True};
 565 observed rows: 1

566 2. **FinBench / Complex Aggregation / medium.**
 567 *Question:* What is the total transferred amount
 568 from accounts owned by person 'Gwar'?

```
569 MATCH (p:Person {personName:  
570 'Gwar'})  
571 -[:OWN_ACCOUNT]->  
572 (src:Account) -[:TRANSFER_TO]->  
573 (:Account)  
574 RETURN DISTINCT SUM(t.amount) AS  
575 TotalTransferredAmount
```

576 *Structure:* join_heavy, aggregation.

577 *Relationships:* OWN_ACCOUNT, TRANS-
 578 FER_TO.

579 *Gates:* RO/Syn/Schema/Exec/Judge.

580 *Result sample:* {TotalTransferredAmount:
 581 14972318.41}; observed rows: 1

582 3. **FinBench / Complex Retrieval / easy.** *Ques-*
 583 *tion:* Which accounts received transfers from
 584 accounts owned by person 'Zof'?

```
585 MATCH (p:Person {personName:  
586 'Zof'})  
587 -[:OWN_ACCOUNT]-> (src:Account)  
588 -[:TRANSFER_TO]-> (dst:Account)  
589 RETURN DISTINCT dst.accountId AS  
590 AccountId, dst.accountType AS  
591 AccountType, dst.isBlocked AS  
592 IsBlocked  
593 LIMIT 300
```

Structure: join_heavy, bounded_result.

Relationships: OWN_ACCOUNT, TRANS-
 FER_TO.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {AccountId:
 4687402787162554854, AccountType:
 credit card, IsBlocked: False}; observed rows:
 3

4. **FinBench / Negation Difference / medium.**
Question: Which accounts owned by person
 'Kant' have not sent any transfers?

```
605 MATCH (p:Person {personName:  
606 'Kant'})  
607 -[:OWN_ACCOUNT]-> (a:Account)  
608 WHERE NOT (a) -[:TRANSFER_TO]->  
609 (:Account)  
610 RETURN DISTINCT a.accountId AS  
611 AccountId, a.accountType AS  
612 AccountType,  
613 a.isBlocked AS IsBlocked  
614 LIMIT 300
```

Structure: join_heavy, negation,
 bounded_result.

Relationships: OWN_ACCOUNT, TRANS-
 FER_TO.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {AccountId:
 4739757132830738341, AccountType:
 merchant account, IsBlocked: False};
 observed rows: 1

5. **FinBench / Path Temporal / medium.** *Ques-*
tion: Which accounts can receive money
 within two transfer hops from accounts owned
 by person 'Sossamon'?

```
628 MATCH (p:Person {personName:  
629 'Sossamon'}) -[:OWN_ACCOUNT]->  
630 (src:Account) -[:TRANSFER_TO*1..2]->  
631 (dst:Account)  
632 RETURN DISTINCT dst.accountId AS  
633 AccountId, dst.accountType AS  
634 AccountType, dst.isBlocked AS  
635 IsBlocked  
636 LIMIT 300
```

Structure: join_heavy, path, bounded_result.

Relationships: OWN_ACCOUNT, TRANS-
 FER_TO.

Gates: RO/Syn/Schema/Exec/Judge.

641	<i>Result sample:</i>	{AccountId:	696
642	4687402787162554854,	AccountType:	697
643	credit card, IsBlocked: False}; observed rows:		698
644	4		
645	6. FinBench / Ranking Topk / medium. Question:		699
646	For accounts owned by person 'Barry',		700
647	which account sent the highest total transfer		701
648	amount?		702
649			703
650	<pre>MATCH (p:Person {personName:</pre>		704
651	<pre>'Barry'})</pre>		705
652	<pre>-[:OWN_ACCOUNT]-></pre>		706
653	<pre>(src:Account)-[t:TRANSFER_TO]-></pre>		707
654	<pre>(:Account)</pre>		708
655	<pre>WITH src, SUM(t.amount) AS</pre>		709
656	<pre>totalAmount</pre>		710
657	<pre>RETURN DISTINCT src.accountId AS</pre>		711
658	<pre>AccountId, src.accountType AS</pre>		
659	<pre>AccountType, src.isBlocked AS</pre>		
660	<pre>IsBlocked,</pre>		
661	<pre>totalAmount</pre>		
662	<pre>ORDER BY totalAmount DESC</pre>		
	<pre>LIMIT 1</pre>		
663	<i>Structure:</i> join_heavy, aggregation, or-		712
664	der_rank, bounded_result.		713
665	<i>Relationships:</i> OWN_ACCOUNT, TRANS-		714
666	FER_TO.		715
667	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.		716
668	<i>Result sample:</i>	{AccountId:	717
669	4732438783436260772,	AccountType:	718
670	internet account, IsBlocked: False}; observed		719
671	rows: 1		720
672	7. FinBench / Simple Aggregation / easy. Question:		721
673	How many accounts are owned by per-		722
674	son 'Kaewsuktae'?		723
675			724
676	<pre>MATCH (p:Person {personName:</pre>		725
677	<pre>'Kaewsuktae'}) -[:OWN_ACCOUNT]-></pre>		726
678	<pre>(a:Account)</pre>		727
679	<pre>RETURN DISTINCT COUNT(DISTINCT a) AS</pre>		
	<pre>AccountCount</pre>		
680	<i>Structure:</i> single_hop, aggregation.		728
681	<i>Relationships:</i> OWN_ACCOUNT.		729
682	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.		730
683	<i>Result sample:</i> {AccountCount: 1}; observed		731
684	rows: 1		732
685	8. FinBench / Simple Retrieval / easy. Question:		733
686	Which accounts are owned by person		734
687	'Barry'?		735
688			736
689	<pre>MATCH (p:Person {personName:</pre>		737
690	<pre>'Barry'})</pre>		738
691	<pre>-[:OWN_ACCOUNT]-> (a:Account)</pre>		739
692	<pre>RETURN DISTINCT a.accountId AS</pre>		740
693	<pre>AccountId, a.accountType AS</pre>		741
694	<pre>AccountType,</pre>		742
695	<pre>a.isBlocked AS IsBlocked</pre>		743
	<pre>LIMIT 300</pre>		
	<i>Structure:</i> join_heavy, bounded_result.		744
	<i>Relationships:</i> CONTAINER_OF,		745
	HAS_MEMBER, HAS_TAG.		746
	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.		747
	<i>Structure:</i> single_hop, bounded_result.		696
	<i>Relationships:</i> OWN_ACCOUNT.		697
	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.		698
	<i>Result sample:</i>	{AccountId:	699
	4732438783436260772,	AccountType:	700
	internet account, IsBlocked: False}; observed		701
	rows: 1		702
	9. SNB / Boolean Existence / medium. Question:		703
	Does person with id 6597069766828		704
	like any post?		705
			706
	<pre>MATCH (p:Person {id:</pre>		707
	<pre>6597069766828})</pre>		708
	<pre>OPTIONAL MATCH (p) -[:LIKES]-></pre>		709
	<pre>(post:Post)</pre>		710
	<pre>RETURN DISTINCT COUNT(post) > 0 AS</pre>		711
	<pre>LikesAnyPost</pre>		
	<i>Structure:</i> single_hop, aggregation, optional.		712
	<i>Relationships:</i> LIKES.		713
	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.		714
	<i>Result sample:</i> {LikesAnyPost: True}; ob-		715
	servd rows: 1		716
	10. SNB / Complex Aggregation / medium. Question:		717
	How many distinct posts		718
	are in forums joined by person with id		719
	6597069766845?		720
			721
	<pre>MATCH (forum:Forum) -[:HAS_MEMBER]-></pre>		722
	<pre>(p:Person {id: 6597069766845})</pre>		723
	<pre>MATCH (forum) -[:CONTAINER_OF]-></pre>		724
	<pre>(post:Post)</pre>		725
	<pre>RETURN DISTINCT COUNT(DISTINCT post)</pre>		726
	<pre>AS</pre>		727
	<pre>JoinedForumPostCount</pre>		
	<i>Structure:</i> join_heavy, aggregation.		728
	<i>Relationships:</i> CONTAINER_OF,		729
	HAS_MEMBER.		730
	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.		731
	<i>Result sample:</i> {JoinedForumPostCount: 1};		732
	observed rows: 1		733
	11. SNB / Complex Retrieval / easy. Question:		734
	Which people are members of forums contain-		735
	ing posts tagged 'Manuel_Noriega'?		736
			737
	<pre>MATCH (forum:Forum) -[:HAS_MEMBER]-></pre>		738
	<pre>(p:Person), (forum)</pre>		739
	<pre>-[:CONTAINER_OF]-></pre>		740
	<pre>(post:Post) -[:HAS_TAG]-> (tag:Tag</pre>		741
	<pre>{name: 'Manuel_Noriega'})</pre>		742
	<pre>RETURN DISTINCT p.id AS PersonId</pre>		743
	<pre>LIMIT 200</pre>		
	<i>Structure:</i> join_heavy, bounded_result.		744
	<i>Relationships:</i> CONTAINER_OF,		745
	HAS_MEMBER, HAS_TAG.		746
	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.		747

Result sample: {PersonId: 4398046511220};
observed rows: 19

12. **SNB / Negation Difference / medium. Question:** Which members of forum 'Wall of Celso Oliveira' have not liked any post?

```
MATCH (forum:Forum {title: 'Wall of
Celso Oliveira'}) -[:HAS_MEMBER]->
(p:Person)
WHERE NOT (p) -[:LIKES]-> (:Post)
RETURN DISTINCT p.id AS PersonId,
p.firstName AS FirstName, p.lastName
AS
LastName
LIMIT 200
```

Structure: join_heavy, negation,
bounded_result.

Relationships: HAS_MEMBER, LIKES.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {FirstName: K., LastName:
Sen, PersonId: 94}; observed rows: 1

13. **SNB / Path Temporal / easy. Question:** Which people are within two knows hops of person with id 4398046511136?

```
MATCH (src:Person {id:
4398046511136})
-[:KNOWS*1..2]-> (dst:Person)
RETURN DISTINCT dst.id AS PersonId,
dst.firstName AS FirstName,
dst.lastName
AS LastName
LIMIT 200
```

Structure: single_hop, path, bounded_result.

Relationships: KNOWS.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {FirstName: Rafael,
LastName: Fernández, PersonId:
4398046511333}; observed rows: 25

14. **SNB / Ranking Topk / medium. Question:** Among forums tagged 'James_K._Polk', which forum has the most members?

```
MATCH (forum:Forum) -[:HAS_TAG]->
(tag:Tag {name: 'James_K._Polk'})
MATCH (forum) -[:HAS_MEMBER]->
(person:Person)
WITH forum, COUNT(DISTINCT person)
AS
memberCount
RETURN DISTINCT forum.title AS
ForumTitle, memberCount
ORDER BY memberCount DESC
LIMIT 1
```

Structure: join_heavy, aggregation, or-
der_rank, bounded_result.

Relationships: HAS_MEMBER, HAS_TAG.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {ForumTitle: Wall of Javed
Khan, memberCount: 33}; observed rows: 1

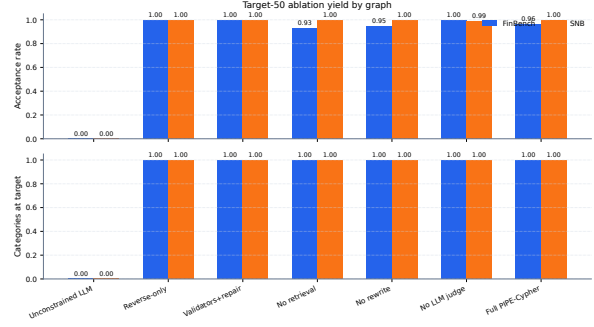


Figure 1: Audited target-50 ablation suite over FinBench and SNB. The suite reaches all graph/variant/category targets and is included as appendix evidence while the larger target-100 and seeded repeat suites continue running for stronger final reliability evidence.

15. **SNB / Simple Aggregation / easy. Question:** How many posts are tagged 'Vietnam'?

```
MATCH (post:Post) -[:HAS_TAG]->
(tag:Tag
{name: 'Vietnam'})
RETURN DISTINCT COUNT(DISTINCT post)
AS
PostCount
```

Structure: single_hop, aggregation.

Relationships: HAS_TAG.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {PostCount: 1}; observed
rows: 1

16. **SNB / Simple Retrieval / easy. Question:** Which post IDs did person with id 4398046511124 like?

```
MATCH (p:Person {id:
4398046511124})
-[:LIKES]-> (post:Post)
RETURN DISTINCT post.id AS PostId
LIMIT 200
```

Structure: single_hop, bounded_result.

Relationships: LIKES.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {PostId: 343597385744}; ob-
served rows: 5

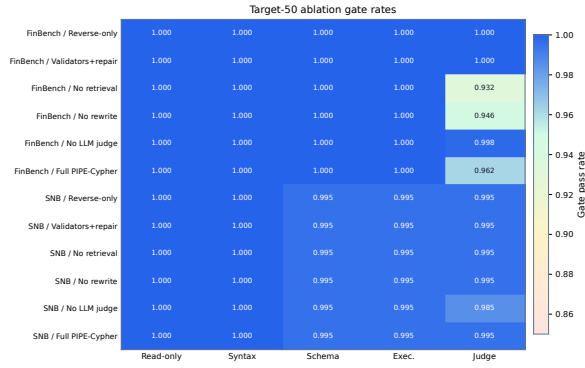


Figure 2: Gate-rate heatmap for the audited target-50 ablation suite. Deterministic read-only and syntax gates are saturated, while schema, execution, and judge rates expose the remaining quality differences across graph and pipeline variants.

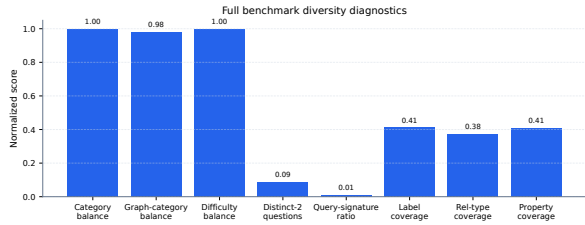


Figure 3: Diversity diagnostics for the full 3,000-example export. Category, graph-category, and difficulty balance are strong by construction; schema coverage and query-signature diversity expose remaining concentration from the fallback seeded-template run.

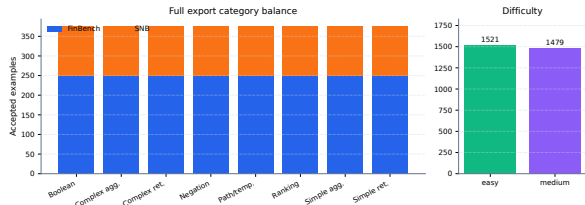


Figure 4: Full 3,000-example export distribution. The benchmark preserves the planned 2:1 FinBench/SNB graph mix while balancing all eight Cypher workload categories and maintaining a near-even easy/medium difficulty split.

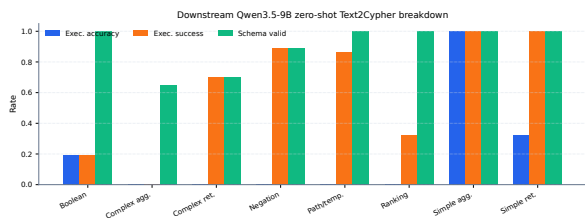


Figure 5: Downstream zero-shot Qwen3.5-9B Text2Cypher evaluation by workload category on the full test split. The breakdown separates final execution accuracy from parse/schema/execution success signals, making difficult operational categories visible.

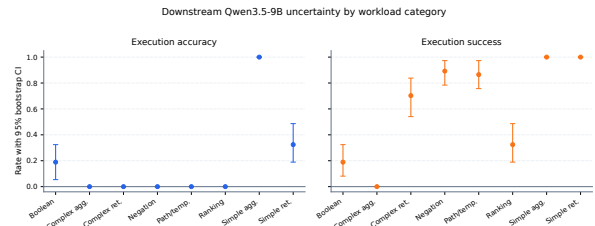


Figure 6: Bootstrap uncertainty for downstream Qwen3.5-9B Text2Cypher evaluation by workload category. Error bars show 95% row-level bootstrap intervals over the full exported test split.